



**R.B.INSTITUTE OF MANAGEMENT
STUDIES(RBIMS) MCA SEMESTER-3**

SUBJECT NAME: : Design and Analysis of Algorithms

SUBJECT CODE: : MC03094011

Assignment-2

Issue Date:21/08/2026 Submission Date:27/08/2026

1.Describe Strassen's algorithm for matrix multiplication and solve following sums.

1. $A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$

$B = \begin{bmatrix} 4 & 0 \\ 1 & 5 \end{bmatrix}$

$ANS = \begin{bmatrix} 9 & 5 \\ 11 & 13 \end{bmatrix}$

2. $A = \begin{bmatrix} 6 & 2 \\ 14 & 10 \end{bmatrix}$

$B = \begin{bmatrix} 3 & 4 \\ 2 & 6 \end{bmatrix}$

$ANS = \begin{bmatrix} 34 & 26 \\ 11 & 7 \end{bmatrix}$

3. $A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$

$B = \begin{bmatrix} 5 & 4 \\ 1 & 3 \end{bmatrix}$

$ANS = \begin{bmatrix} 24 & 12 \\ 1 & 3 \end{bmatrix}$

4. $A = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$

$B = \begin{bmatrix} 1 & 5 \\ 2 & 6 \end{bmatrix}$

$ANS = \begin{bmatrix} 19 & 41 \\ 5 & 1 \end{bmatrix}$

5. $A = \begin{bmatrix} 2 & 8 \\ 5 & 1 \end{bmatrix}$

$B = \begin{bmatrix} 3 & 4 \\ 2 & 6 \end{bmatrix}$

$ANS = \begin{bmatrix} 54 & 64 \\ 21 & 27 \end{bmatrix}$

6. $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$

$B = \begin{bmatrix} 5 & 1 \\ 6 & 7 \end{bmatrix}$

$ANS = \begin{bmatrix} 54 & 64 \\ 21 & 27 \end{bmatrix}$

$ANS = \begin{bmatrix} 54 & 64 \\ 21 & 27 \end{bmatrix}$

2.Calculate the expected number of hires for 7 candidates using indicator random variables.

3.Define randomized algorithms and explain types.

4.Explain the substitution method for solving recurrences.

5.Explain Recursion Tree Method with example.

6.Describe the master method for solving recurrences.

7. Explain maximum sub-array and 3 techniques.

• Solve sums using Kaden's algorithm, Divide and Conquer and brute force techniques

1.Array: [4, -1, 2, -7, 5, 2, -1, 3]

Maximum Sum: 9

Subarray: [5, 2, -1, 3]

2.Array: [-3, 8, -2, 4, -5, 6, -1]

Maximum Sum: 11

Subarray: [8, -2, 4, -5, 6]

3.Array: [10, -3, -4, 7, 6, 5, -4, -1]

Maximum Sum: 21

Subarray: [7, 6, 5, -4, -1]

4.Array:[-2, -5, 6, -2, -3, 1, 5, -6]

Maximum Sum: 7

Subarray: [6, -2, -3, 1, 5]

5.Array:[3, -2, 5, -1, 6, -3, 2, -5, 4]

Maximum Sum: 11

Subarray: [3, -2, 5, -1, 6]

6.Array:[-8, 4, -1, 2, 1, -5, 4]

Maximum Sum: 6

Subarray: [4, -1, 2, 1]

7.Array:[2, -1, 2, 3, 4, -5]

Maximum Sum: 10

Subarray: [2, -1, 2, 3, 4]

8.Array:[-1, 4, -2, 5, -5, 2, 2, -1, 3]

Maximum Sum: 7

Subarray: [4, -2, 5]

9.Array:[5, -9, 6, -2, 3, -1, 5, -6]

Maximum Sum: 11

Subarray: [6, -2, 3, -1, 5]

10.Array:[-2, 3, -1, 3, -2, 4, -5

] Maximum Sum: 7

Subarray: [3, -1, 3, -2, 4]